

24- λ Coherent Optical Communications Using Passive Coherence Cloning of Dark-Soliton Microcombs for Homodyne Detection

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Abstract We report on the first demonstration of coherent multi-wavelength transmission using passive coherence cloning of Kerr microcombs. Our scheme leverages a coherence-cloned replica of the transmitter comb as a multi-wavelength local oscillator for homodyne reception of 24 16QAM and 32QAM channels with reduced digital-signal-processing effort. ©2026 The Author(s)

Introduction

Offering a multitude of highly stable phase-locked optical tones spaced by tens to hundreds of GHz, Kerr microcombs are key to a wide range of emerging applications, including massively parallel wavelength-division multiplexing (WDM) [1–3], high-precision optical ranging and LiDAR [4,5], and ultra-broadband photonic-electronic signal processing [6–9]. So far, these demonstrations have mainly leveraged the intrinsic coherence of the various tones within a single comb, guaranteeing perfectly equidistant WDM channels or ensuring mutual coherence between different frequency bands in ultra-broadband signal generation and measurement [6–9]. Equally interesting but much less explored is the concept of establishing and leveraging coherence between different microcombs, e.g., at the transmitter (Tx) and the receiver (Rx) of an optical link. A first approach relied on active coherence cloning of two microcombs, using a shared pump laser along with electronic control of the free spectral range (FSR) of the Rx comb [10]. This has led to a WDM transmission experiment with 20 channels, each carrying 21 GBd 16QAM signals, offering an aggregate line rate of 1.68 Tbit/s. However, while confirming that coherence cloning can significantly reduce the complexity of digital frequency-offset compensation (FOC) at the Rx, the approach still required active control of the receiver comb along with conventional intradyne detection due to a residual FSR offset. Moreover, the demonstration relied on conventional bright solitons, exhibiting rather low pump-to-comb power-conversion efficiency – one of the key shortcomings of microcombs in optical communications. On the other hand, passive coherence-cloning techniques have been demonstrated, obviating electronic feedback loops and instead establishing robust and ultra-precise coherence via nonlinear optical interactions [11–13]. The underlying approaches comprise sharing of the continuous-wave (CW) pump along with optical injection of a seed tone [13] or the entire comb [11,12] and have been demonstrated for both bright [11,13] and dark [12] solitons. However, these demonstrations were not geared towards communication or signal-processing applications. The potential of passively coherence-cloned microcombs in optical communications hence still remains to be explored.

In this paper, we report on the first demonstration of

multi-wavelength coherent transmission using passive coherence cloning of Kerr microcombs. Our demonstration relies on a pair of optically packaged dark-soliton Kerr-comb generators, leveraging advanced 3D-printed facet-attached microlenses (FaML) [14,15] for efficient fiber-chip interfaces. Passive coherence cloning is achieved by injecting two unmodulated tones of the Tx comb into the Rx comb generator — without any electronic feedback loop. In the coherence-cloned regime, the Rx comb is an exact copy of its Tx counterpart, i.e., comb-line frequencies and phases are identical by construction, making the Rx comb a perfectly stable multi-wavelength local oscillator (LO) for parallel homodyne detection. Interestingly, our concept fully obviates the need for FOC, thereby greatly simplifying the entire digital-signal-processing (DSP) chain. We demonstrate the viability of our scheme in a proof-of-concept experiment using 24 WDM channels, each carrying an 80 GBd 16QAM or 32QAM signal. We transmit the signals over an 85 km-long fiber, reaching an aggregate net data rate of 8.3 Tbit/s. To the best of our knowledge, our experiment represents not only the first demonstration of Kerr-comb-based multi-wavelength homodyne reception, but also the first use of passive coherence cloning of Kerr microcombs in optical communications.

System concept and experimental setup

Figure 1a shows the concept and setup of our multi-wavelength transmission demonstration leveraging passive coherence cloning of dark-soliton microcombs for parallel homodyne detection. The Tx comb, indicated by a dashed blue box in Fig. 1a, relies on an optically packaged high-Q Si_3N_4 racetrack micro-ring resonator (MRR, intrinsic $Q_i > 10^7$) [16] with an FSR of $f_{\text{FSR}} = 44.6782$ GHz and normal group-velocity dispersion (GVD). Fiber arrays (FA) couple to the Si_3N_4 chip by 3D-printed facet-attached microlenses (FaML) [14,15]. The fiber-chip coupling loss is below 1.3 dB for both the Tx comb and Rx comb modules. A dual-tone pump, generated by a continuous-wave (CW) external-cavity laser (ECL, $\lambda = 1555.45$ nm) with a subsequent Mach-Zehnder modulator (MZM) and an erbium-doped fiber amplifier (EDFA), is injected into the Tx MRR. The MZM is biased at the null transmission point and driven by an RF synthesizer (RF syn.) with a frequency f_{RF} close to half of the FSR of the Tx MRR,

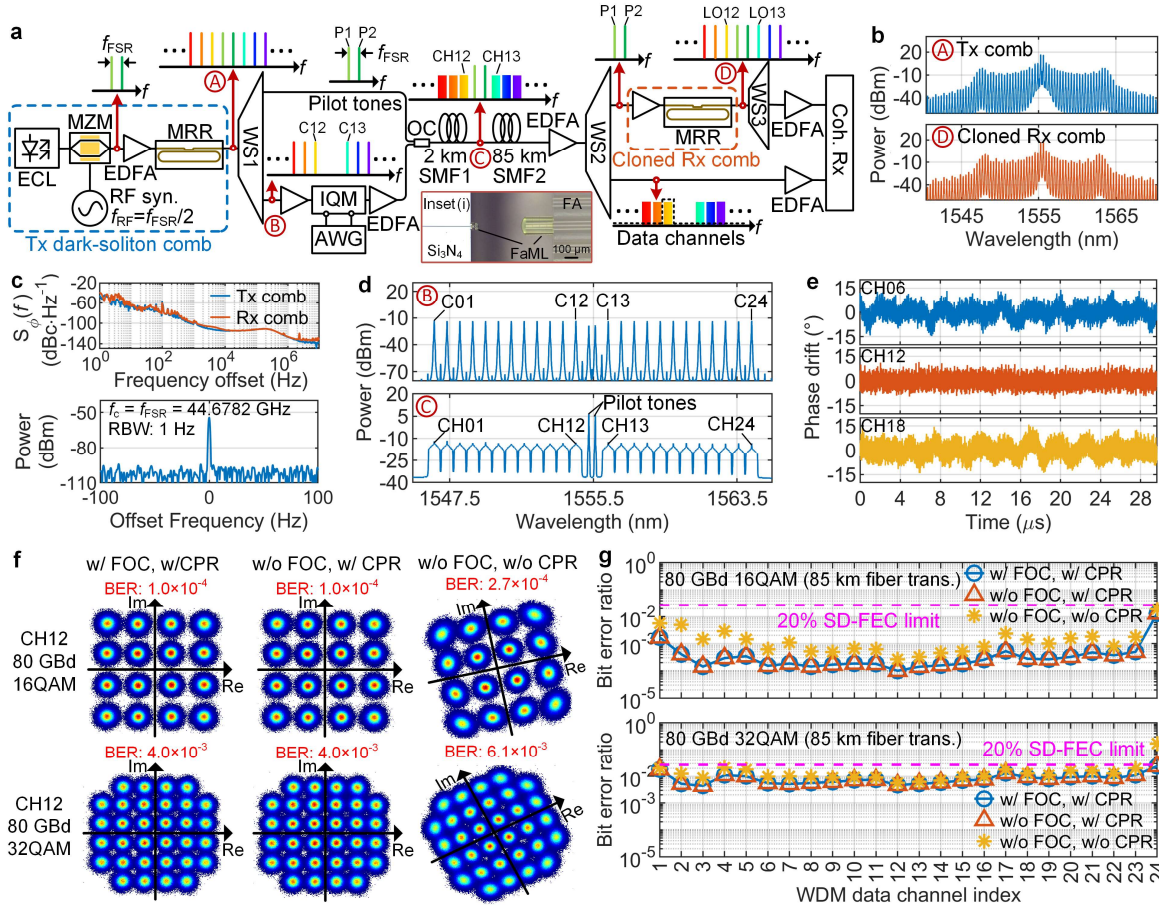


Fig. 1: Experimental setup and results. (a) Concept and setup of our WDM transmission experiment using passive coherence cloning of dark-soliton microcombs for parallel homodyne detection. The dark-soliton comb generator at the transmitter (Tx, dashed blue box) comprises an optically packaged high-Q racetrack Si_3N_4 micro-ring resonator (MRR) with normal group-velocity dispersion (GVD) and a free spectral range of $f_{\text{FSR}} = 44.6782$ GHz. Fiber-chip coupling is accomplished by facet-attached micro-lenses (FaML), 3D-printed to the chip and the fiber array (FA), see Inset (i). A dual-tone pump, generated by an external-cavity laser (ECL), a subsequent Mach-Zehnder modulator (MZM), and an erbium-doped fiber amplifier (EDFA), is coupled to the MRR. The MZM is biased at a null transmission point and driven by an RF synthesizer (RF syn.) such that the first-order modulation sidebands of the pump tone coincide with a pair of cavity resonances, thereby igniting a dark-soliton Kerr comb. A wave-shaper (WS1) separates two residual pump tones (pilot tones) and 24 carrier tones. The carriers are divided into two groups (C01 ... C12 and C13 ... C24) with a line spacing of $2 \times f_{\text{FSR}} = 89.3564$ GHz, which are sent to an in-phase/quadrature modulator (IQM). The IQM is driven by an arbitrary-waveform generator (AWG, Keysight M8194A). The pilot tones are combined with the modulated signal through an optical coupler (OC). A 2 km-long dispersive single-mode fiber (SMF1) decorrelates adjacent data channels. The combined signal is then transmitted over an 85 km-long fiber link (SMF2). At the receiver (Rx), a second WS (WS2) separates the pilot tones and extracts a data channel of interest. The pilot tones are amplified by an EDFA and used to pump a second MRR (Rx MRR) with an FSR close to the one of the Tx MRR, thereby generating a coherence-cloned Rx comb (dashed orange box). A third WS (WS3) selects a comb tone to serve as a local oscillator (LO) for homodyne reception of the channel of interest in the subsequent coherent receiver (Coh. Rx). (b) Optical spectra of the Tx comb (Top, Point A in Subfigure a) and coherence-cloned Rx comb (Bottom, Point B in Subfigure a). (c) Measured power spectral density (PSD) of RF phase noise of the repetition signals of the Tx and Rx combs (Top), and electrical spectrum of the RF beat note between a Tx comb tone and the subsequent tone taken from the Rx comb. The resolution bandwidth (RBW) is 1 Hz and the center frequency of the RF beat note f_c is exactly the same as the individual comb f_{FSR} , i.e., $f_c = f_{\text{FSR}} = 44.6782$ GHz. (d) Top: Optical spectrum of the power-flattened carrier tones (C01 ... C24, Point B in Subfigure a). Bottom: Optical spectrum of the combined pilot tones and the 24 wavelength-multiplexed data signals (CH01 ... CH24, Point C in Subfigure a). (e) Carrier phase drift at the Rx for three exemplary channels (CH06, CH12, CH18), confirming the stability of the coherence cloning scheme. (f) Exemplary constellation diagrams of 80 GBd 16QAM and 32QAM signals for three cases related to frequency-offset compensation (FOC) and carrier-phase recovery (CPR) in the receiver DSP chain. (g) Measured bit error ratios (BER) for 24 channels with 80 GBd 16QAM (Top) and 32QAM (Bottom) transmission after an 85 km-long fiber (SMF2). The blue circles, orange triangles, and yellow asterisks represent the measured BER with three different cases with (w/) and without (w/o) FOC and CPR in the receiver DSP chain. No BER penalty is to be seen when FOC is omitted, and the BER remains below the threshold for soft-decision forward-error-correction (SD-FEC) with 20 % overhead.

i.e., $f_{\text{RF}} = f_{\text{FSR}}/2 = 22.3391$ GHz. This leads to dominant first-order sidebands which coincide with a pair of adjacent cavity resonances and which ignite a dark-soliton Kerr comb via cascaded four-wave mixing (FWM) [8], see top panel of Fig. 1b for the associated comb spectrum at Point A in Fig. 1a. The Tx comb features 48 tones with a power per line above -10 dBm. We then use a wave shaper (WS) to select

the two residual pump tones (pilot tones, P1 and P2), which serve as a pump for passive comb-cloning at the receiver, and 24 carrier tones (C01 ... C24), where C01 ... C12 are to the left and C13 ... C24 are to the right of the pilot tones. Note that we selected only every second comb tone as a carrier such that the carriers within each group are spaced by $2 \times f_{\text{FSR}} = 89.3564$ GHz. The carriers are

modulated by an in-phase/quadrature modulator (IQM), which is driven by an arbitrary-waveform generator (AWG, Keysight M8194A). The resulting multi-channel data signals are combined with the two pilot tones via an optical coupler (OC). A subsequent 2 km-long dispersive single-mode fiber (SMF1) decorrelates the data signals in neighboring WDM channels to emulate realistic multi-wavelength transmission conditions. The signals are then sent over an 85 km-long standard single-mode fiber (SMF2). At the receiver, a second WS (WS2) extracts a certain wavelength channel CH_μ , $\mu = 01 \dots 24$, and simultaneously separates the two pilot tones P1 and P2 from the overall WDM signal. The pilot tones are then amplified by an EDFA and used as a pump for the Rx MRR, offering normal GVD and an FSR close to its Tx counterpart, see dashed orange box in Fig. 1a. Driven by essentially the same pump tones as the Tx MRR, the Rx MRR generates a coherence clone of the Tx comb, see bottom panel of Fig. 1b for the optical spectrum of the generated Rx comb as measured at Point ② in Fig. 1a. The resulting Rx comb is then sent to a third WS (WS3) that extracts the comb tone LO_μ associated with WDM channel CH_μ of interest. The comb tone is then routed to a subsequent coherent receiver (Coh. Rx) to serve as an LO for homodyne detection, exploiting the fact that the comb tone LO_μ and the carrier of the associated WDM channel CH_μ are perfectly coherent. The detected IQ signals are captured by a high-speed real-time oscilloscope (RTO, Keysight UXR0804A) and processed by an offline DSP. In a first step, we verify the coherence between the Tx and the Rx combs by measuring the repetition rate f_{FSR} and the RF phase noise of the repetition signals, see Fig. 1c. We find that the two combs feature exactly the same FSR and the same RF phase noise. In a second step, we measured the RF beat note of a Tx comb tone with the subsequent tone taken from the Rx comb, finding that the two tones are exactly spaced by f_{FSR} , see bottom panel of Fig. 1c for the electrical spectrum with a resolution bandwidth (RBW) of 1 Hz. The RF beat note has a center frequency of $f_c = f_{\text{FSR}} = 44.6782$ GHz and features a narrow-band RF spectrum, indicating strong mutual coherence between the two combs.

Coherent WDM transmission demonstration

Using the setup shown in the previous section, we finally performed a WDM transmission experiment, using 80 GBd 16QAM or 32QAM signals on an overall 24 channels. At the Tx, the selected comb lines for each channel are equalized in power by the WS (WS1). The resulting optical spectrum of the carrier tones for 24 channels ($\text{CH}01 \dots \text{CH}24$) is measured at Point ③ in Fig. 1a and shown in the top panel of Fig. 1d. The bottom panel shows the optical spectrum of the corresponding WDM signal, combined with the two pilot tones, as measured at Point ④ in Fig. 1a. To quantify the residual phase fluctuations between the received signals and the locally generated LO tones, we select three exemplary channels ($\text{CH}06$, $\text{CH}12$, and $\text{CH}18$) and evaluate the carrier phase drifts using the carrier-phase recovery (CPR) algorithms in the DSP chain, see Fig. 1e. All

channels exhibit a phase drift within $\pm 15^\circ$ over an observation time of 29 μs . To verify that our scheme actually reduces the DSP effort for FOC, we evaluate both 80 GBd 16QAM and 32QAM signals with (w/) and without (w/o) FOC and CPR in the Rx DSP chain. The exemplary constellation diagrams for $\text{CH}12$ are shown in Fig. 1f. Activating CPR and w/o FOC (Fig. 1f, middle column) yields essentially no bit error ratio (BER) penalty compared to the case when FOC and CPR are both activated (Fig. 1f, left column), confirming the notion that FOC can be completely omitted, thereby reducing the DSP effort. We also evaluate the BER of the received signal without adaptive CPR, compensating only the constant phase rotation of the constellations for the entire block of 2^{23} symbols, see the right column of Fig. 1f, where only a small BER penalty is observed. Figure 1g shows the measured BER for all data channels with 80 GBd 16QAM (top) and 32QAM (bottom) for the different cases. Omitting the FOC does not lead to any change, and the BER of all channels (orange triangles and blue circles) remains below the soft-decision forward-error-correction (SD-FEC) limit with 20% overhead (2.7×10^{-2} , magenta dashed lines) [17]. Replacing the adaptive CPR with a simple phase rotation of symbols within the recorded block leads to only a slight BER penalty, see yellow asterisks in Fig. 1g. Except for channel 24, the BER for all channels is also below the SD-FEC limit with 20% overhead, with again reduced DSP effort. The larger BER penalty for channel 24 is due to the relatively low optical power of the corresponding LO tone. We also estimate the overall aggregate net transmission capacity, calculated from the normalized generalized mutual information (NGMI) for each channel with an appropriately selected FEC code rate [18]. This yields a net transmission capacity of 8.3 Tbit/s at an aggregate line rate of 9.6 Tbit/s, using 32QAM on all channels. To the best of our knowledge, our experiment represents not only the first demonstration of Kerr-comb-based multi-wavelength homodyne reception, but also the first use of passively coherence-cloned Kerr microcombs in optical communications.

Summary

We demonstrated the first multi-wavelength homodyne transmission using passive coherence cloning of Kerr microcombs. Using a pair of optically packaged dark-soliton comb generators, passive coherence cloning is achieved by injecting two unmodulated tones from the Tx comb into the Rx microresonator without any electronic feedback loop, thereby generating an exact copy of the Tx comb as the local oscillator (LO) for parallel homodyne detection. Our scheme fully obviates the need for digital frequency-offset compensation and greatly simplifies the entire DSP chain. We demonstrate the viability of our scheme in a WDM experiment, transmitting 24 channels over an 85 km-long fiber, each channel carrying an 80 GBd 16QAM or 32QAM signal. We reach an aggregate net data rate of 8.3 Tbit/s. To the best of our knowledge, our work represents the first demonstration of multi-wavelength homodyne reception using passive coherence cloning of microcombs.

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